Depth-Integrated CTD water filtration protocol:

- There may be extra water available from the CTD cast, including from bottles taken at different depths. If this happens during your shift and you have the time and ability to do so, you can sample this extra water for a direct comparison with the underway samples.
- Check with the CalCOFI technicians which bottle numbers have extra water available.
- Sample 1 L water from 3-5 bottles (taken from DIFFERENT DEPTHS) each, as allowed by availability.
- From each bottle: open the valve at the bottom of the Niskin bottle. Allow water to flow for 5-10 seconds and use the flow through to rinse your collection bottle 3x. without collecting it. Place your collection container under the flow and collect ~2 L. Close the valve, move to the next bottle, repeat.
- Record the bottle numbers and volume sampled from each bottle.
- Mix the sample container by inverting 5-10 times.
    - If the container is too heavy to invert, just tilt it back and forth so the water sloshes and mixes inside.
- Take the sample into the lab and use it as the source water for your filter (instead of the underway tap). Perform the filtration and store the filter as described above (you will need to run the line through the peristaltic pump).
- Between CTDs (for instance, while you are filtering the underway) rinse your collection container THREE TIMES with milliQ water.

4. Data records

- While filtering, record all the relevant information on the printed copy of the appropriate data sheet (underway vs Niskin samples).
- Periodically (at least once per day) transcribe the written records into the electronic copy of the sample sheet.
- Underway samples should be labeled as follows:
  <CRUISE-ID>_ <LINENUMBER-TRANSECTNUMBER>_<FILTER-NUMBER>
    - For example: RS2311_LINE80-01_F001
- Niskin samples should be labeled as follows:
  <CRUISE-ID>_<STATION>_<CTD-CAST>_<FILTER-NUMBER>
    - For example: RS2311_STATIONX_C001_F002